50.021

Real vs AI Generated Images

Yong Jun Han (1009942)

Contents

AI image generators are easily accessible to the public today. From Midjourney to Deepfakes,

These images can be used to spread a false narrative, or to swindle.

Case Study

The “Brad Pitt” Incident

A French woman was scammed by a fraudster using AI generated images.

The woman was contacted by someone claiming to be Brad Pitt’s mother, and later by someone claiming to be the actor.

“Brad Pitt” then sent her photos of him in a hospital bed, successfully duping her.

Aftermath: £700,000 and a broken marriage

Dataset

CIFAKE: Real and AI-Generated Synthetic Images (Kaggle)

120,000 Files (.jpg)

- test
  - fake (50000)
  - real (50000)
- train
  - fake (10000)
  - real (10000)

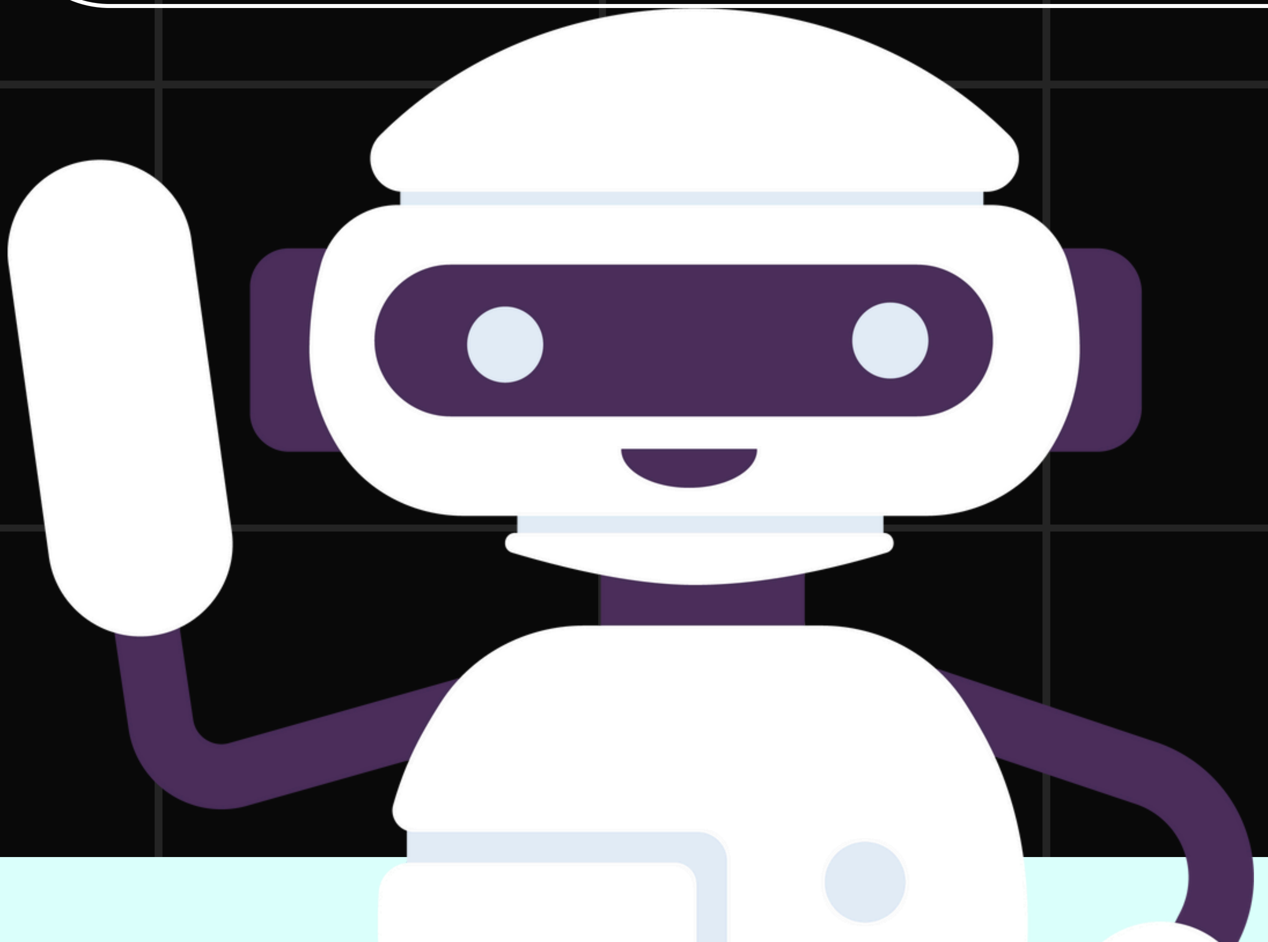
Existing Works

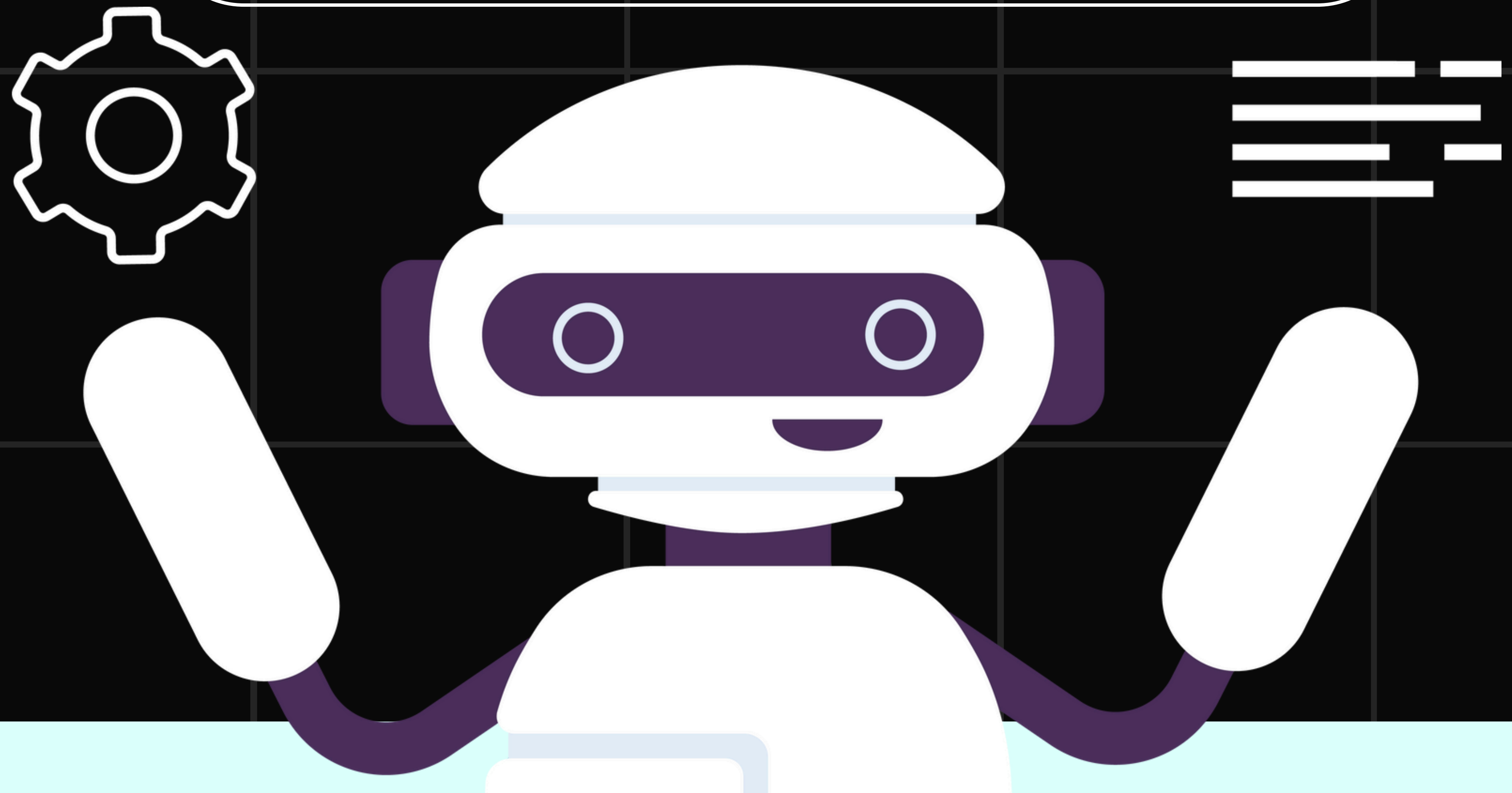
VGG

- CNN
- Deep architecture, higher accuracy.

MobileNet

- CNN
- More lightweight, efficient.





Preliminary Ideation

Preprocessing

- Rescaling of train images, dividing 255.
- Dataset split into train and valdiation sets (80:20) ratio.
- Ensure image sizes are uniform (32x32).

Preliminary Ideation

Model to be Used: CNN

Data Preprocessing

↓

Model Setup & Training

↓

Evaluation & Benchmarking

Results

Evaluation

Self-built Model

- Accuracy: 0.837
- Loss: 0.363
- Training Speed: 377s
- Test Accuracy: 0.7710
- Test Speed: 16s

MobileNet v2

- Accuracy: 0.9074
- Loss: 0.2291
- Training Speed: 1,954s
- Test Accuracy: 0.9074
- Test Speed: 130s

GUI Demonstration

[Link]

Thank you!